Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force and motion

Meaning in my words How does a net force change motion?	**How can you tell?** A cart feels a net force right. What else tells you whether it is speeding up right now?
Example or sketch Sketch a cart moving left while a net force points right. Which way does acceleration point?	**Non-example and why** “The force points right, so the cart is already going right.” What evidence is missing?

Interval

Meaning in my words How would you explain interval to a friend?	**How can you tell?** The clock reads 3 s at the start and 7 s at the end. How long is the interval?
Example or sketch You are using velocities from 3 s and 7 s. Which strip of the graph belongs in your calculation?	**Non-example and why** A partner uses 7 s as the elapsed time. Which part did they include by accident?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Reversal

Meaning in my words How would you explain reversal to a friend?	**How can you tell?** On a velocity–time graph, how can you tell a cart actually reversed?
Example or sketch A cart goes from +2 m/s through zero to −1 m/s. Draw its direction before and after.	**Non-example and why** A bus stops, then keeps going east. Does stopping prove it reversed?

Area

Meaning in my words How would you explain area to a friend?	**How can you tell?** What do you do with the area below the time axis when you find displacement?
Example or sketch Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show?	**Non-example and why** Someone labels the area −2 m/s. Which graph reading did they use?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

force | acceleration | velocity | position

My marks mean

Can we release a forecast?

20

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Identify which measurements a position forecast needs.

CHOOSE ONE JOB TOGETHER

MISSION SYSTEMS ENGINEER ☐

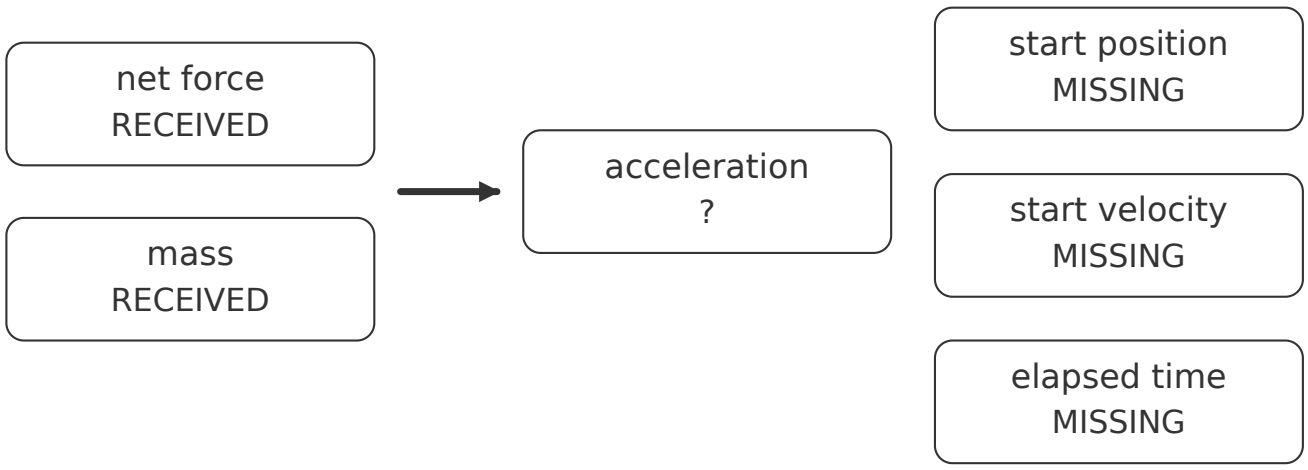
Mark what you can calculate now. Send the telescope operator a request for the measurements still needed.

NEWS ASSIGNMENT EDITOR ☐

Prepare a 'known / still needed' graphic. Tell the anchor which result can air and which must wait.

Point to a clue. Keep your first marks; add the update using the color key.

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.



01 OUR FIRST CALL

What can the desk release now? What measurements must the crew still collect?

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Is the forecast ready after the update? Say what you can release and what is still missing.

Switch speakers. If XJ reverses direction during the forecast, how would that affect a report of distance versus displacement?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks